# Matched Source Supplement

Dale James Raymond
ORCID: 0009-0001-1682-8529

29 October 2026 | Version 1.0
DOI: 10.5281/zenodo.23752103

**Abstract**

This supplement identifies the mathematical data and source-to-kernel interfaces of the main manuscript. Its Theorem B.1 derives all 67 physical source chords directly from primitive payments. The finite replay has 15 actual stages, including an additional exact source-reconstruction consumer. The numerical endpoint, source data, kernel plan and 221,010 comparison arcs are preserved. This edition supersedes referee edition 3 of 28 October 2026; the numerical plan, source table and comparison history are unchanged; Appendix B adds the direct source reconstruction.

## 1 Manuscript and numerical result

The matching main manuscript is `Unconditional_0107965.pdf`, with its identically named TeX source. Its computer-assisted upper bound is

$$\Lambda \le B_* := \frac{427693072077942473661578098491}{396140812571321687967719751680 0} < 0.107964910079.$$

The initial strip comes from OpenAI's ordinary-zeta nonvanishing theorem for $\operatorname{Re} s > 7/8$, formalised in Lean at the exact revision identified in the main paper. The functional equation and the normalisation $H_0(z) = \xi((1+iz)/2)/8$ give squared halfwidth $9/16$. The strict initial seed is $562500000001/10^{12}$. The external theorem, written analytic comparison and finite certificate consumers are separate parts of the argument. This package does not claim a new local full Lean build or an end-to-end Lean formalisation.

## 2 One source formula for every row

For the scalar signed nodes $(h_{ij}, a_{ij}, w_{ij})$, put

$$\mathcal{L}_i S_t(x) = \sum_j \frac{w_{ij}}{2}\{S_t(x + a_{ij}, h_{ij}) + S_t(x - a_{ij}, h_{ij})\}, \qquad S_t = -\operatorname{Im}(H_t'/H_t).$$

A zero shift denotes one evaluation. On the row's closed thermal cell $I_i = [t_i^-, t_i^+]$, set $\theta = (t - t_i^-)/(t_i^+ - t_i^-)$. Theorem B.1 gives the source of the actual final functional as

$$J_i(t,x) \ge M_i - N_i - (1-\theta)A_i - \theta B_i \qquad (t \in I_i,\ x \ge 6000000185827).$$

Its four fields are reconstructed by

$$M_i = \alpha_i m_i + 2m_i^{\mathrm{j}}, \qquad N_i = \alpha_i(D_i - \rho_i) + 2D_i^{\mathrm{j}},$$
$$A_i = \alpha_i(H_{i,-} - C_i) + 2H_{i,-}^{\mathrm{j}}, \qquad B_i = \alpha_i(H_{i,+} - C_i) + 2H_{i,+}^{\mathrm{j}}.$$

Here $m_i$ is the independently bounded scalar main term, $D_i$ is the complete primitive native debit, $\rho_i$ is its legitimate physical refund, $H_{i,\pm}$ are full raw finite heads and $C_i$ is the complete simultaneously paid arithmetic credit. The superscript j denotes one jointly bounded derivative functional. Outside rows $0, 16, 44$, the jet terms vanish and $\alpha_i = 1$.

For the 39 retained rows, $C_i = C_i^{\mathrm{comp}} + d_i$, where $d_i$ is the total replacement common-clock payment. For the 28 shared rows, $C_i = \Delta_i^{\mathrm{fix}} + \mu_i/(5000000) + C_i^{\mathrm{orbit}}$, with the fixed lower-capacity branch explicitly required. The shared capacity budget, residual prime capacities and alternative near/far main payments are proved in the main manuscript's arithmetic appendix. These payments are not inferred from the historical name or status of an intermediate table.

## 3 The extra mathematical bindings

The final `direct-source` consumer makes the following checks explicit in addition to the primitive inequalities checked in the other 14 stages.

1. Every actual scalar main field satisfies $m_i \le \frac{1}{2}(\sum_j w_{ij}) \log 690988$. A rational logarithm enclosure proves this inequality without accepting an upstream main field.
2. Each intermediate source adapter preserves the fields and complete functional that its primitive proof establishes. The two row families cover all 67 rows exactly once.
3. The final plan's derivative multiplier, centre and every coefficient agree exactly with the proposal whose source bound is proved. The orders on rows $0, 16, 44$ are respectively $4, 4, 8$. Row 44 uses the replacement order-eight jet once.
4. The order-eight spectral certificate contains one value enclosure for the left endpoint and one for every isolated critical root. The list length is checked before corresponding entries are compared.
5. Direct evaluation gives all 268 final source-field identities, with matching scalar nodes, thermal cells and contact intervals.

The supplied values satisfy these extra checks. The new consumer uses raw witness data and exact arithmetic; it does not read a historical PASS flag as an acceptance premise. Its finite assembly check relies on the analytic identification and the separately executed feasibility consumers for the inequalities being assembled.

## 4 Physical and geometric obligations

The common finite model uses one actual central cutoff, physical phase and normalisation. The value and derivative errors arise from the same analytic discrepancy. The Gaussian majorant uses 11/12, the Euler derivative term is retained, and only the variable physical charge is refunded. Fixed coordinate and normaliser terms remain charged.

The kernel certificates concern the same signed functional. They cover continuous spacing and contact-height domains; harmonic extension gives the corresponding inequality for every other zero height. The entire signed own-pair term is subtracted once. The stationary wall has corrected physical floor 697/900 on its complete declared domain. The finite-head, positive-time compactness and repeated-contact arguments are in the main manuscript and are not consequences of a checksum test.

The exact flow consumer checks every one of the 221,010 arcs, their positive ideal curves, source charges, own pair, domain caps, outward rounding and joins. At the terminal time,

$$T = \frac{22101}{204800}, \qquad Q_T = \frac{790237796383053674715864 3}{2^{96}} > 0.$$

The final de Bruijn contraction adds $Q_T/2$, giving $B_* = T + Q_T/2$. The separate consequences note proves only deductions from this terminal strip and does not replace the proof of the strip itself.

## 5 Exact data identities and provenance

The following paths are relative to `Proof_and_Replay/Lambda_Peer_Review/`. Preserved referee-edition-3 files are under `Proof_and_Replay/Release_Support/reconstruction/`; the new run records are under `Proof_and_Replay/Release_Support/fresh_replay/`.

**Operational source** `inputs/workspace/lambda_round15/flow_main_v2/MAIN_CLOCK_SOURCE.json`
`b59905d7b994d3424ba14f9347be75a2429ae4d4b0de2c2404971ffda1d4d536`

**Final functional plan** `inputs/workspace/lambda_round14/material/m8_late/M8_LATE_PLAN.json`
`ec346b7c1136eda38021eb6323033b3b48ed21e6f7458eeeac7cc9579db2497a`

**Full comparison history** `inputs/workspace/lambda_round16/ordinary_7_8_flow/full_flow/FLOW.json.gz`
`a664e829a3681d96e474bcf875d80cba7b16fe8f65e050c69f01b5ea07de4737`

**New direct source consumer** `verification/direct_source/verify_direct_source.py`
`a0e8cca86919957d82b714413adc4cc967a32aee94c8b521554d6d2a8a01ce19`

The current release manifest identifies the matching main PDF and TeX, all supplements and both archives. The two historical Prior PDFs are byte-preserved from the earlier 0.116398 deposit. Their historical statements do not supersede the current source theorem or verifier.